

Max Marchionda

Minneapolis, MN · (608) 498-5604 · maxmarchionda@gmail.com

EXPERIENCE

Sr. Principal Software Engineer — Machine Learning

Optum (UnitedHealth Group) · Minneapolis, MN · June 2017 – Present

Sr. Principal Software Engineer Feb 2024 – Present

Principal Software Engineer July 2022 – Feb 2024

Senior Software Engineer Feb 2021 – July 2022

Software Engineer II June 2019 – Feb 2021

Data Scientist II June 2018 – June 2019

Data Scientist June 2017 – June 2018

- Led and built an AI-powered benefits planning application that leverages machine learning to optimize health benefit configurations, surfacing recommendations through a complex business logic grid evaluated on both a plan and market basis — enabling more strategic, data-driven benefit decision-making at scale.
- Designed and led development of a custom web application that processes competitor ad data using a multi-modal agentic AI workflow, analyzing thousands of ads monthly to automatically surface unique insights — saving over \$2M in annual operational costs.
- Architected and built an internal AI-powered chat experience enabling employees to query 25,000+ developer and data documentation pages, supporting 1,500+ active users and streamlining access to critical technical knowledge.
- Led a cross-functional team in designing and deploying a model discovery and consumption platform at UHG, accelerating AI adoption across 15+ business units and enabling tracking of 450+ AI projects through the full ML lifecycle.
- Contributed to the design and implementation of a scalable enterprise ML development platform, optimizing model development workflows and infrastructure for production-grade deployment.
- Developed and deployed end-to-end training and scoring pipelines for predicting 20+ chronic and acute conditions (including Atrial Fibrillation and Diabetes) using healthcare claims and clinical data.

EDUCATION

B.S. Computer Science · B.S. Chemistry

University of Minnesota · Minneapolis, MN · Sept 2012 – May 2017 · GPA: 3.40

SKILLS

Cloud Platforms: Google Cloud Platform (GCP), Microsoft Azure, AWS (Lambda, API Gateway, S3, Route 53, CloudFront)

AI & Machine Learning: Large Language Models (LLMs), Agentic AI Workflows, Python, Azure AI Search, Azure OpenAI, Gemini, ChatGPT, ADA v2

Full-Stack Development: React, Next.js, Node.js, Figma

E-Commerce & Marketing: Shopify, third-party marketing integrations, targeted advertising, customer retention tooling

ACTIVITIES & INTERESTS

Photography · Music · Collectibles · Cooking · App Development · Video Games